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Asset tokenization in plain English

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Abstract

Asset tokenization is an expansion of blockchain technology that allows digital assets to be bought, sold, and traded on blockchains. Digitizing real assets will help create universally accessible, fast, liquid, and transparent investment and financial systems. However, blockchain and related asset tokenization are still in their infancy, and there are obstacles that need to be overcome before widespread adoption occurs. This article describes what it means to tokenize an asset on a blockchain, how tokenization is accomplished, and why it has the potential to disrupt a variety of industries, particularly the financial and real estate industries. Challenges that are associated with implementing asset tokenization are also explored.

KEYWORDS

asset tokenization, benefits of tokenization, blockchain technology, challenges of asset tokenization

1 | INTRODUCTION

Blockchain technology, which gained prominence through the launch of the cryptocurrency, bitcoin, is being expanded to many other business processes. One such expansion of blockchain use is the tokenization of assets. Tokenization is a method that converts rights to an asset into digital tokens that can be bought, sold, and traded on blockchains. Tokenization on blockchains has been a steady trend since early 2018. It seems that everything is being tokenized from paintings, diamonds and company stocks to luxury boats, jetliners, and real estate. There are even projects aiming to tokenize oceans, celestial objects, and other “exotic assets.”

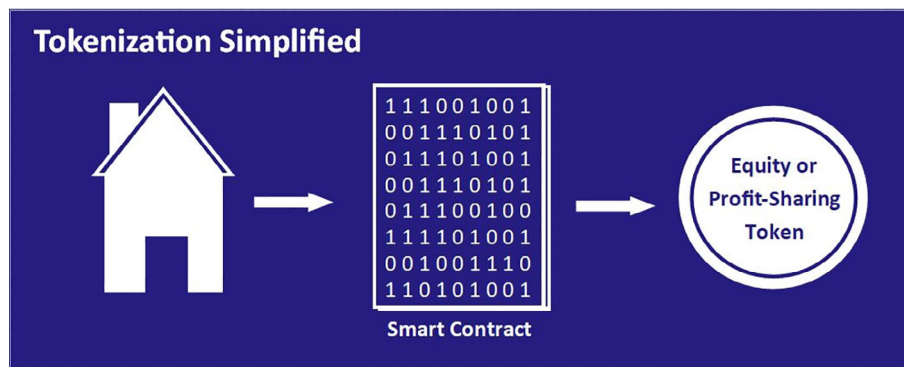
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2 | HOW DOES ASSET TOKENIZATION WORK?

To “digitize” means to turn non-digital into digital (e.g., you take a video shot on tape and turn into DVD). In the context of asset tokenization, physical assets are turned into digital assets. Then, the digital assets can be subdivided and the subunits can be represented by a digital token. Exhibit 1 illustrates this process. All tokenization is digitization but not all digitization is tokenization. Tokenization is a special form of digitization that supports fractional investment and ownership.

Representing real assets as digital tokens allow issuers and holders of the tokens to achieve the benefits of cryptocurrency, that is, security, liquidity, and immutability, to real-world assets. An example of the benefits of tokenization can be seen in the ability to “fractionalize ownership” of real assets. Imagine you want to invest in real estate but your initial investment is modest—say \$5,000. Perhaps you want to start small and increase your investment gradually. For instance, you decide to invest a

EXHIBIT 1 Tokenization simplified



couple 1,000 dollars every 3 or 4 months. Obviously, in a traditional real estate market, this is quite difficult to do. How does someone buy two- or three-square meters of an apartment?

Now assume you are the owner of the apartment and you need cash quickly. The apartment is valued at \$200,000, but you only need \$10,000. How can you use your equity in the apartment to obtain \$10,000 quickly and easily? Traditional methods would require a loan from a financial institution that is both labor and time-intensive. Tokenization can transform this apartment into 200,000 digital tokens (the number is arbitrary; we could have issued 2 million tokens). Each token, representing a 0.0005% share of the apartment could then be easily traded on a variety of blockchains with smart contracts (computer programs where the terms of an agreement are written directly into lines of code). When you buy one token, you actually buy 0.0005% of the ownership in the apartment. Buy 100,000 tokens and you own 50% of the apartment. Buy all 200,000 tokens and you are 100% owner of the apartment. Because blockchain is a distributed public ledger that is immutable, once you buy tokens, no one can “erase” your investment.

3 | BENEFITS OF TOKENIZATION

In addition to facilitating fractional ownership as described above, asset tokenization offers the following benefits.

1. *Global access to investments:* People in advanced economies have easy access to vast investment opportunities. In the USA, at least 50% of families hold stocks in publicly traded companies (Bricker et al., 2017). The figure is dramatically less for economies in transition and developing economies with only 1 or 2% of these
2. *Greater liquidity:* Fractionalized ownership and universal access to investment opportunities provided by tokenized assets support greater liquidity for the underlying assets. In addition to allowing smaller investment amounts and having access to significantly more markets, digital assets are simply easier to trade. It is much easier to trade digital tokens on a blockchain than to trade real estate, ships, or diamonds in their physical form.
3. *Reduced need for intermediaries:* With blockchain technology, there is less need for middlemen. Tokens can be traded directly between buyers and sellers without banks, lawyers, or other intermediaries. However, periodically, intermediaries may be required to verify the existence and condition of the underlying asset.
4. *Faster transactions:* On blockchains, assets can change hands at the speed of thought. There are no forms to complete, papers to sign, waiting periods, and so forth. Most blockchain platforms can be accessed from a web browser or a mobile device, and transactions can be executed in real-time.
5. *Transparency:* Blockchains, via the anonymous cryptocurrencies they support, have been associated with criminals, terrorists, and con men. In reality, blockchain is one of the most transparent financial systems ever invented. In fact, every single transaction can be publicly traded on the digital ledgers, making them much more transparent than a traditional banking system.
6. *Immutability:* Immutability refers to the fact that once a block of transactions is validated and added to a blockchain, the information is almost impossible to

alter or delete. This is because the chain of blocks and the transactions they contain are replicated on computers that are literally located throughout the world. If a bad actor tries to alter a transaction, all the other computer operators (called miners) would know and would not allow the alteration. This feature makes transaction history not only transparent, but secure.

4 | EXAMPLES OF TOKENIZED ASSETS

The previous example of the apartment was completely arbitrary. In reality, practically any asset, tangible or intangible, can be tokenized on a blockchain. Assets such as precious metals and stones, artwork, luxury boats, airplanes, intellectual property, company equity, real estate investment trusts (REITs), debt and equity investment funds, over-the-counter derivatives, and customer loyalty points are prime candidates for tokenization.

5 | TOKENIZING EQUITY: STOKR

STOKR is a peer-to-peer interface that allows small and medium-size companies to seek and obtain funding from both professional and retail investors. Tokens issued on STOKR's interface qualify as transferable securities according to European capital market rules, and as such comply with European transparency, disclosure, and investors' protection requirements.

STOKR tokenizes a venture's distributable profits because each token issued on STOKR represents a fixed percentage of the distributable future profits of the ventures.

5.1 | How does it work?

Ventures can issue European Union-compliant security tokens to raise capital in return for a specified share of their profit. Thus, STOKR allows firms to get funding with no loss of equity or independence. For investors, STORK is an ideal platform as well. Anyone can become a venture capitalist on STORK. From mobility to fashion, fully vetted ventures are available for investing.

6 | TOKENIZING INTELLECTUAL PROPERTY: BREAKER

Breaker, the blockchain content distributor is formerly known as SingularDTV, empowers artists and creators to transact directly with their consumers and audience. The

system created by Breaker runs on Ethereum and relies on a tokenized Smart Contract System called SNGLS. These SNGLS tokens represent a share in the intellectual property (IP) of Breaker; a total 1 billion SNGLS have been issued 50% of which was made available to public, 40% is held in a storage vault for continuous reinvestment into the platform, and the 10% remainder was distributed amongst the main investors of Breaker.

6.1 | How does it work?

Content creators create tokens in the Breaker platform. Tokens represent a project. Tokens can be distributed to different parties that collaborate with the content creator (screenwriter, director, actors, band members, etc.) or sold directly to consumers.

As tokens are programmable smart contracts, content creators can program reward-sharing properties into the token. This means that the entire reward generated by a project through its distribution in the Breaker ecosystem can be directly claimed by token holders in amounts proportional to the number of tokens they own with no middleman or regulating authority involved.

7 | ART AND COLLECTIBLES: CODEX

The biggest drivers of an item's monetary value are its identity and chain of ownership, also known as provenance. Provenance is important for collectibles, artwork, and even classic cars. Provenance information still relies on paper documents. These documents are hard to share, easy to fake, and require constant re-verification. This is a problem that CODEX solves with its Codex Protocol for creators, auction houses, and collectors worldwide by tokenizing provenance information.

7.1 | How does it work?

Any information regarding the identity or history of an item can be uploaded to a Codex Record. This includes digital receipts, photographs of receipts, certificates of authentication, and so forth. Such Blockchain transactions of the Record will appear at the bottom of the Record under "provenance."

Blockchain transactions are used to document transactions when the underlying asset's ownership is changed and thus CODEX allows the artist to "stay in touch" with the work. As these transactions are immutable, the creation of fakes and forgeries becomes practically impossible.

Selling artworks with blockchain records enables tracking of resale royalty payments.

8 | STEPS TO ASSET TOKENIZATION

It may sound easy. Just “tokenize” your assets! But, of course, tokenization is much easier said than done. Below is a brief description of the steps necessary to accomplish asset tokenization.

1. *Asset identification*: The first step in the asset tokenization process is to identify the asset that you plan to tokenize. The asset should be fully defined and the nature and amount of asset you want to trade on a blockchain should be completely described.
2. *Asset evaluation*: Most assets will require assessment and valuation from a reputable third party. Normally, this is done by appraisers and/or auditing firms. The assessment and valuation typically become part of a whitepaper or other document used as marketing material for the sale of the tokens.
3. *Tokenomics*: Once the asset is identified and evaluated, token parameters must be determined in order to create the actual digital tokens. Decisions such as how many tokens will be issued, whether and how new tokens will be added (i.e., rate of inflation), what happens to lost tokens, what happens to unsold tokens, etc., must be made. This step is arguably the most difficult in the process. Many tokenization projects fail at this stage.
4. *Smart contract creation and audit*: Once token economics are decided, actual tokens are created through the development of smart contracts. A prerequisite of smart contract development is the selection of the specific digital ledger (blockchain) to be used to sell the tokens. There are many blockchains to choose from, each with different features and technical requirements. Once the platform is selected, a smart contract developer and auditor should be engaged. Smart contract audits are a critical part of the process, because auditors can identify coding problems before the contract is deployed. Once the contract is live on a blockchain, it cannot be altered.

9 | TYPES OF ASSET-BACKED TOKENS

There are two broad types of tokens used for real assets: equity tokens and profit-sharing tokens (often called revenue-sharing or dividend tokens). Which token is most appropriate depends on the business model. Tokenization

of certain assets is more natural with equity tokens while other assets are more suited for profit-sharing tokens. Token classification is also important from a regulatory perspective. Depending on the characteristics of tokens, as well as how and to whom they are sold, regulators may determine that the sale constitutes a sale of securities and, therefore, demand that the issuer meet strenuous regulatory criteria before offering the tokens for sale.

Suppose the asset to be tokenized is a \$25 million hotel that generates \$1 million in annual net profit. Because many tokens are freely traded on exchanges, the value at any point is determined by market forces. However, equity tokens generally gain value as the valuation of the underlying asset increases. Investing in equity tokens is similar to investing in a traditional stock of a company. If 1 million tokens were sold to represent 100% of the value of the hotel, each token would represent 0.0001% equity of the hotel. (The issuer could choose to peg the 1 million tokens to 50% of equity or issue 10 million tokens pegged to just 10% of the equity in the hotel. This is one of the decisions made during the “tokenomics” phase of the tokenization process.)

Profit-sharing tokens work differently. Profit-sharing tokens are pegged to a certain percentage of the asset's profits. In the hypothetical hotel example, the 1 million tokens could be pegged to say 10% of the net profits generated by the hotel. Thus, 1 million tokens would be worth \$100,000 or 10 cents per token. The token will gain value if the hotel's profits go up and lose value if the profits go down. However, unlike equity tokens, profit-sharing token holders receive actual profits without having to sell their tokens.

The exact circumstances under which profit-sharing token holders would receive a portion of the tokenized asset's profit and how much would be received depends on the terms of the agreement when the tokens were created and sold. Those terms are programmed into the token (smart contract) and immutably stored on a blockchain. The process of agreeing to terms and then following them each year when profit is determined is very similar to the issuance of preferred stock. What is the applicable percentage? Is the stock participating or nonparticipating? Convertible? Callable? Voting? All of these terms are decided before the stock is sold and becomes a contract with the shareholders. The profit-sharing terms in an asset tokenization arrangement are also predetermined and are coded into an immutable blockchain.

10 | CHALLENGES TO ASSET TOKENIZATION

Businesses that wish to take advantage of asset tokenization need to overcome several challenges as described below.

10.1 | Legal challenges

While technology presents several challenges, the principal challenge comes from government and industry regulation, or more specifically, from the lack of such regulation. While regulators in many countries are beginning to consider rules for trading tokens and cryptocurrencies, to date, no country has well-developed guidance for such transactions. This is particularly problematic should a dispute arise about asset ownership. For example, what happens if the company that holds the underlying real asset sells the property? What if the asset is stolen or damaged? Do token owners have a legal claim against the holder of the physical asset? Who is responsible for storage, insurance, and so forth? And, even more fundamentally, if token holders reside in 25 different countries, all of which are different from the country in which the asset is stored, which country has jurisdiction? Obviously, legal changes are needed to accommodate these new business models.

Fixing the legal and regulatory problems would necessitate some sort of centralization. The whole idea of blockchain and smart contracts is to create an environment where intermediaries, especially central governments, are not needed or utilized. While this is possible to achieve when tokenizing digital assets (e.g., photos, music, video and other multimedia content, vouchers, digital collectibles, and so forth), when tokenizing physical assets, this is not the case. When physical assets are involved, there will have to be a certain amount of centralization and involvement of intermediaries for tokenization to work.

10.2 | Business partners

Businesses wishing to implement asset tokenization need to find reliable technical enablers, financial partners, and legal advisors who are well versed in blockchain, smart contracts, and digital tokens. As the market grows, these business partners will be more available and experienced. Today, demand is straining supply.

10.3 | Compliance

As discussed above, token sales and trading may require compliance with securities regulations. In addition, companies involved in asset tokenization may be subject to a number of compliance frameworks and processes like AML,² KYC,³ and MiFID.⁴ Compliance can be very costly. For instance, it is estimated that preparations for MiFID II (successor of MiFID) cost firms an estimated \$2.1 billion each (Regulatory Compliance: MiFid II Solutions, n.d.). While the cost of compliance can be high, the

cost of noncompliance can be even higher. Injunctions, disgorgement orders, legal defense costs, loss of reputation, and so forth can present a significant risk of issuing organizations.

10.4 | Cybersecurity

Blockchain-related businesses are extremely vulnerable to cybersecurity threats. In 2018 alone, there were over 100 security incidents involving \$2 billion in stolen cryptocurrency (Blockchain Graveyard, n.d.). Cybersecurity is a complex field that is even more complex in a globally distributed blockchain system. This is because every system has points of attack. These points, often called “attack vectors,” are points where a malicious user can extract from or change data in the environment. The sum of all attack vectors makes the attack surface. Generally, the larger the attack surface, the more vulnerable the system. (This is obvious given hitting a larger target is generally easier than hitting a smaller target).

Blockchain's attack surface is enormous. In addition to being globally distributed, blockchain can be attacked at a data layer or a network layer, which are low-level attacks. At a higher level, risks come from consensus and smart contract layers and finally from the application layer.

11 | CONCLUSION

Asset tokenization is truly a transformative and disrupting paradigm. Digitizing real assets will help create universally accessible, fast, liquid, and transparent investment and financial systems. However, blockchain and related asset tokenization are still in their infancy, and there are obstacles that need to be overcome before widespread adoption occurs. The good news is that progress is being rapidly achieved. Many governments realize that blockchain is a game-changer and that the sooner they develop enabling legislation, the greater the competitive advantage that jurisdiction will realize.

ENDNOTES

¹ National Securities Depository Limited (National Securities Depository Limited, n.d.) and Central Depository Services Limited (Central Depository Services (India) Limited, n.d.), two largest providers of special investment accounts in India list approximately 30 million investment accounts or approximately 2% of the population.

² A set of laws, regulations, and procedures intended to prevent criminals from disguising illegally obtained funds as legitimate income.

- ³ Know your customer, alternatively known as know your client, is the process through which businesses verify the identity of their clients to assess their suitability, along with the potential risks of illegal intentions towards the business relationship.
- ⁴ The Markets in Financial Instruments Directive 2004/39/EC is a European Union law that provides harmonized regulation for investment services across the 31 member states of the European Economic Area (the 28 EU member states plus Iceland, Norway and Liechtenstein). The directive's main objectives are to increase competition and investor protection in investment services (Regulatory Compliance: MiFid II Solutions, n.d.).

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